DASC 3240, L10

Scatter plot and regression – Assignments

2023/24 SPRING SEMESTER

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Today’s outline

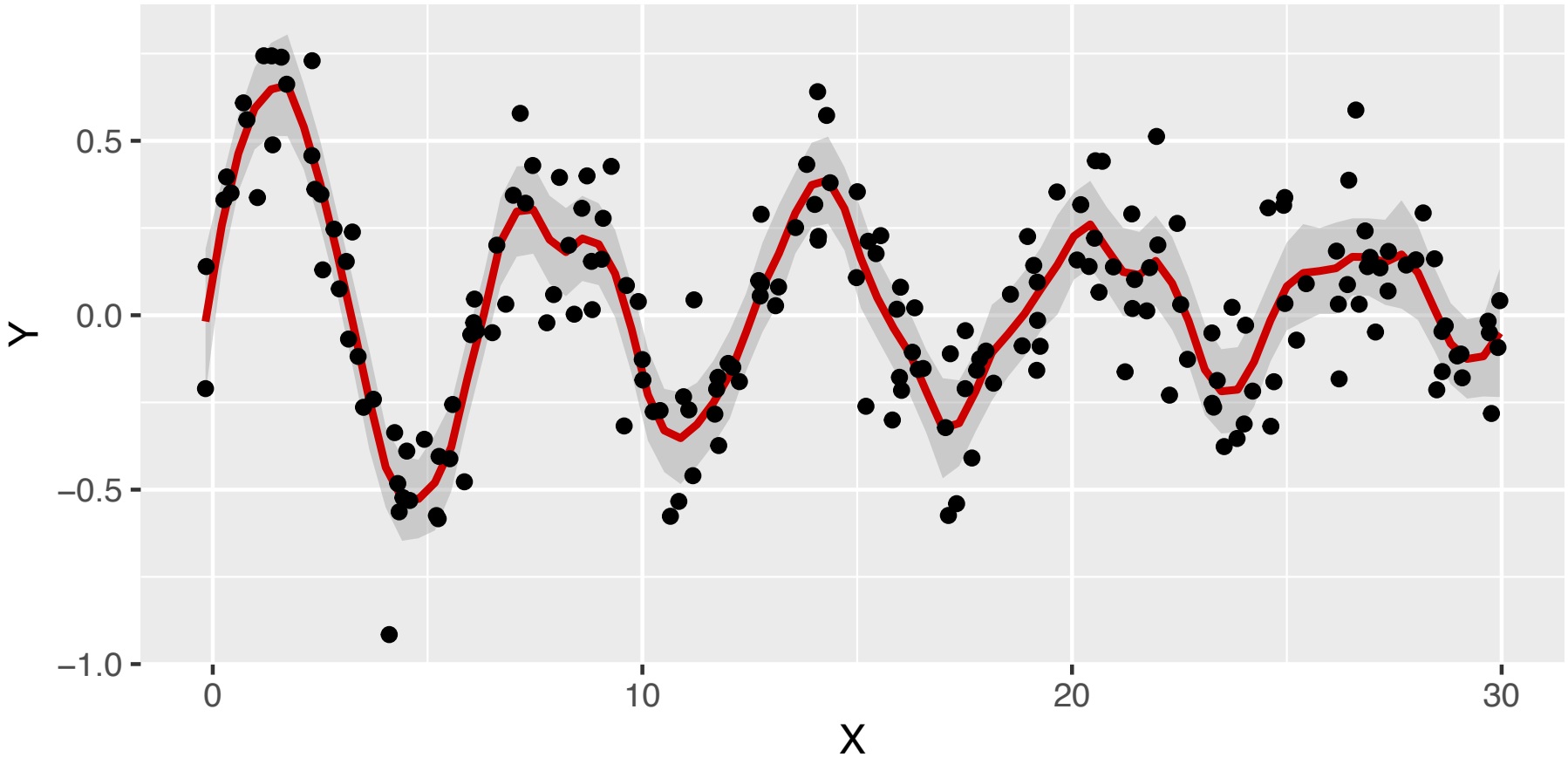
- Scatter plot and regression – Assignments
  - Data
  - Today’s assignments

Data

Custom dataset | d_nonlinear.csv

- Numeric data that show a nonlinear relationship

https://github.com/ong8181/DASC3240/blob/main/data/d_nonlinear.csv



Today’s assignments

SCATTERPLOT AND REGRESSION

Assignments

- **d_nonlinear.csv**
  1. Linear regression: Load d_nonlinser.csv and draw a scatter plot of X and Y. Then, add a linear regression line without confidence intervals.
  2. LOWESS: Load d_nonlinser.csv and draw a scatter plot of X and Y. Then, add a LOWESS fitting curve without confidence intervals.
  3. Visualize the effect of the parameter frac on LOWESS (see detailed instructions in the jupyter notebook file)

  4. **[Optional]** In Task 3, you generate 6 plots. How can we choose the most appropriate LOWESS curve? Please explain your thoughts with supporting evidence (e.g., book, url, and/or scientific manuscript). You can use any AI tool(s), but note that such tool(s) often provide wrong reference(s).

Example figures

- **d_nonlinear.csv**
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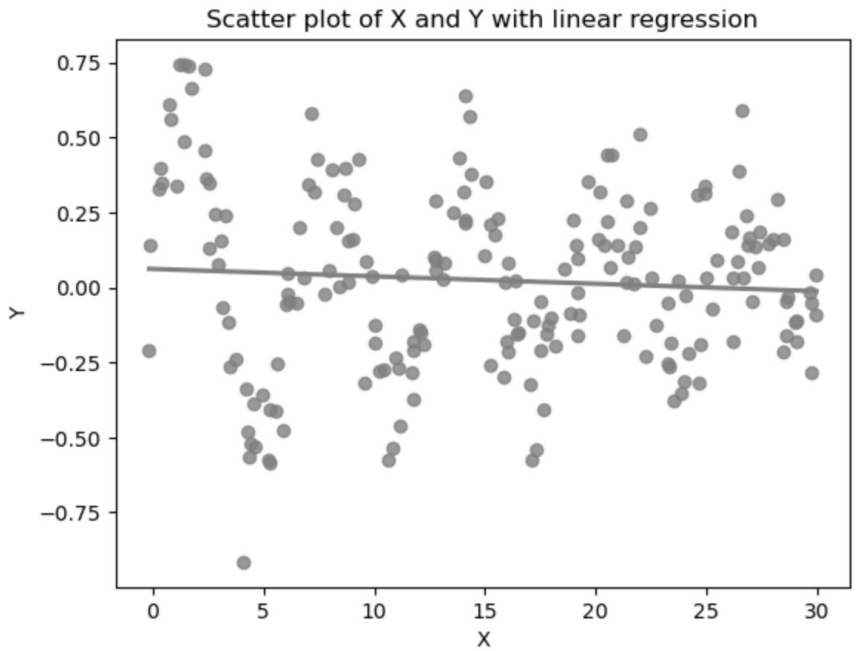

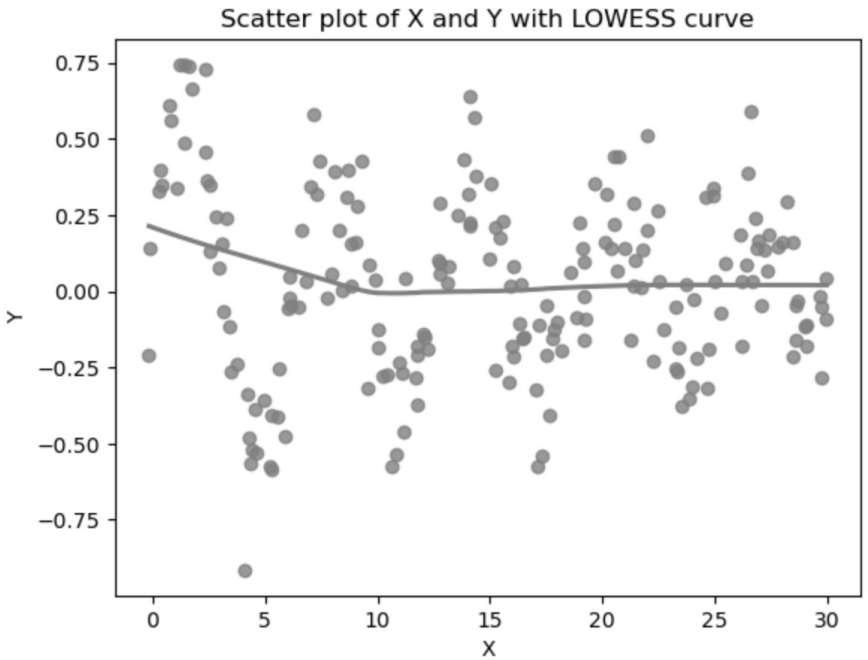

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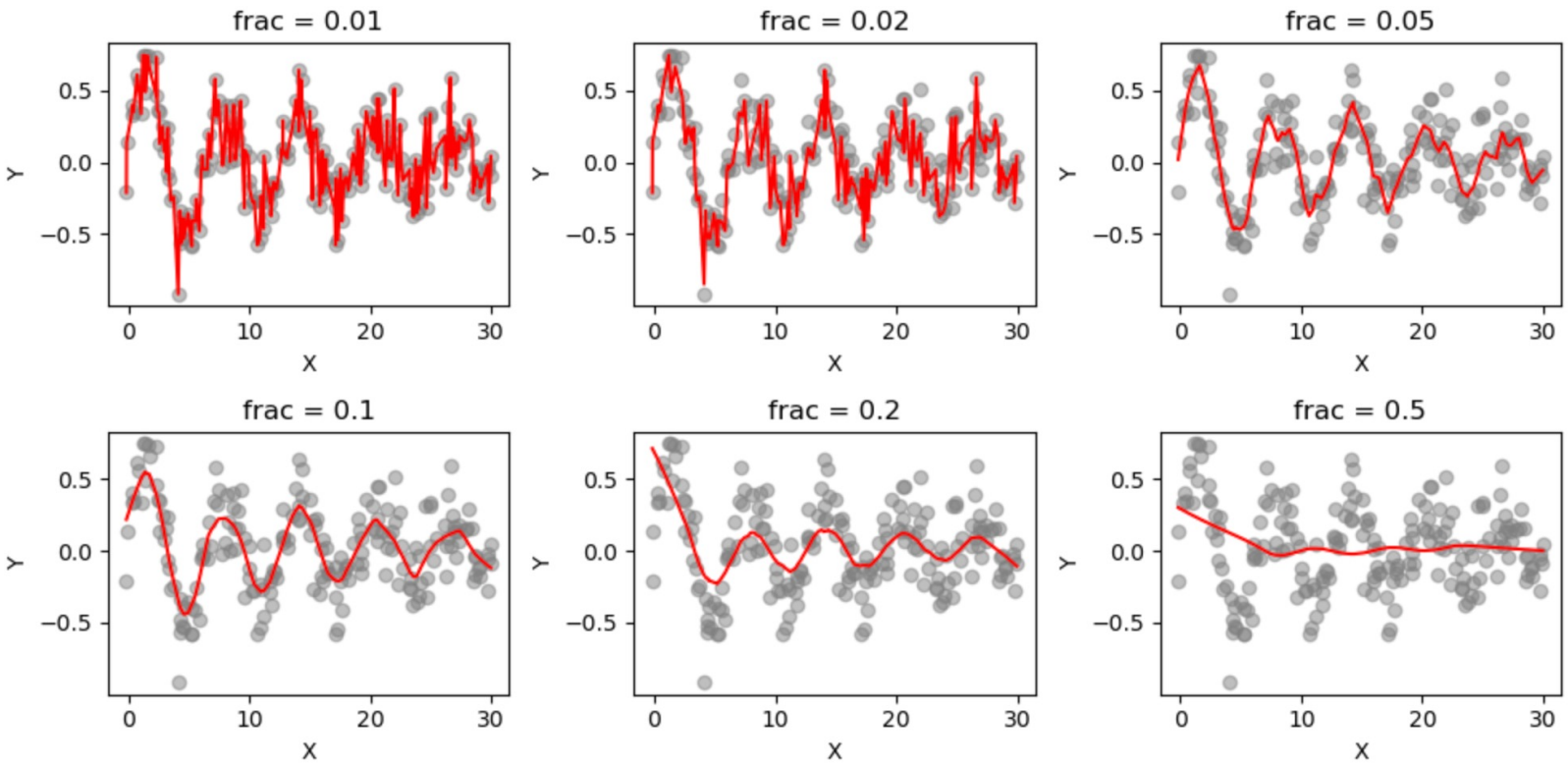
3. Visualize the effect of the parameter frac on LOWESS (see detailed instructions in the jupyter notebook file)